

Smart Classroom On-Site Assembly Checklist

This checklist is for group members assembling the demo hardware on-site.

Current build note: no capacitors are installed.

A. Modules From the 48-Piece Kit Photo

Photo No.	Module	Current Use	Interface
#4	Traffic Light Module	Used	R -> UNO D10, Y -> UNO D11, G -> UNO D12, GND -> UNO GND
#5	Active Buzzer Module	Used	S -> UNO D7, + -> UNO 5V, - -> UNO GND
#16	Flame Sensor	Used	VCC -> UNO 5V, GND -> UNO GND, DO -> UNO D8, AO -> UNO A3
#18	PIR Motion Sensor	Used	VCC -> UNO 5V, GND -> UNO GND, OUT -> UNO D2
#25	Analog Gas Sensor	Used	VCC -> UNO 5V, GND -> UNO GND, AO -> UNO A1, DO not connected
#30	TEMT6000 Light Sensor	Used	VCC -> UNO 5V, GND -> UNO GND, SIG -> UNO A0
#39	LM35 Temperature Sensor	Used	VCC -> ESP32 5V/VIN, GND -> ESP32 GND, OUT -> ESP32 GPIO34
#38	5V Single Relay Module	Not used	replaced by external 6-port relay module
#7	Digital Push Button Module	Not used	current build uses a bare 4-leg tactile button

B. Extra Components Not From the Photo

Component	Interface
KS0001 Keystudio UNO R3	main edge controller
ESP32 dev board	MQTT gateway and LM35 ADC reader
6-port relay module	DC+ -> UNO 5V, DC- -> UNO GND, IN -> UNO D6
12V 4-wire fan	Red -> relay NO/ON, Black -> 12V adapter -, Yellow -> UNO D3, Blue -> UNO D9
12V 2A adapter	+ -> relay COM, - -> fan black and common ground
discrete classroom LED	UNO D5 -> 220 ohm resistor -> LED long leg, LED short leg -> GND
emergency 4-leg button	one side -> UNO D4, opposite side -> UNO GND, code uses INPUT_PULLUP
UART divider	UNO A5 -> 10k -> ESP32 GPIO16; ESP32 GPIO16 -> 20k -> GND
ESP32 command TX	ESP32 GPIO17 -> UNO A4
common ground	UNO GND -> ESP32 GND -> 12V adapter -
ADC pulldown	ESP32 GPIO34 -> 100k -> GND
dummy ADC pulldown	ESP32 GPIO35 -> 100k -> GND

C. UNO Pin Table

UNO Pin	Connected Device
D2	PIR OUT
D3	fan yellow tach
D4	emergency button to GND
D5	classroom LED through 220 ohm
D6	relay module IN
D7	active buzzer S
D8	flame sensor DO
D9	fan blue PWM
D10	traffic light R
D11	traffic light Y
D12	traffic light G
A0	TEMT6000 SIG
A1	gas sensor AO
A2	legacy UNO LM35 input, currently not main temperature source
A3	flame sensor AO
A4	SoftwareSerial RX from ESP32 GPIO17
A5	SoftwareSerial TX to ESP32 GPIO16 through divider

D. ESP32 Pin Table

ESP32 Pin	Connected Device
GPIO16 RX2	receives UNO A5 telemetry through 10k/20k divider
GPIO17 TX2	sends commands to UNO A4
GPIO34 ADC1	LM35 OUT
GPIO35 ADC1	dummy reference channel with 100k pulldown to GND
5V/VIN	LM35 VCC
GND	LM35 GND, UNO GND, 12V adapter - common ground

E. Power-On Order

Recommended for stable LM35 readings:

1. Connect all grounds first. 2. Power ESP32 and verify dashboard/gateway is online. 3. Power UNO. 4. Power 12V adapter for fan. 5. Wait 10-20 seconds for telemetry and temperature filter to stabilize.

F. Expected Dashboard Signals

Item	Expected
ESP32 status	online
UNO telemetry	updates roughly once per second
temperature source	ESP32 after stable LM35 ADC
safety alarm	0 during normal mode
status light	green for normal/lighting, yellow for cooling/energy-saving, red for safety alarm
command ACK	updates after dashboard/Qwen/manual commands

G. Common Assembly Mistakes

Symptom	Check
ESP32 receives no UNO data	A5 divider to GPIO16, common GND, UNO firmware flashed
ESP32 command not reaching UNO	GPIO17 -> UNO A4, no accidental connection to GPIO16 divider midpoint
temperature too high	LM35 VCC should be ESP32 5V/VIN, OUT to GPIO34, GPIO34/GPIO35 pulldowns present
temperature floats with sensor unplugged	GPIO34 is high impedance; check 100k pulldown
fan not spinning	12V adapter + -> relay COM, relay NO/ON -> fan red, fan black -> 12V -
fan always spinning	relay trigger polarity or wrong NO/NC terminal
tach is zero	fan yellow -> UNO D3, common ground, tach pullup if needed
red/yellow/green wrong	swap R/Y/G wires on D10/D11/D12
button always pressed	button must straddle breadboard gap; D4 uses INPUT_PULLUP
MQTT offline	laptop IP/MQTT host mismatch or hotspot/router changed